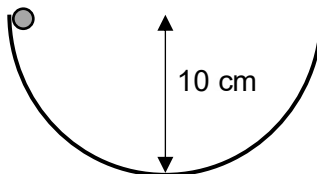


- 9** A small object of mass 0.050 kg is released from rest at the rim of a heavy, smooth semi-spherical bowl of radius 10 cm as shown in the figure below.



When the object passes the bottom of the bowl, what is the normal force exerted on it by the bowl?

**A** 0.49 N

**B** 0.98 N

**C** 1.5 N

**D** 2.0 N

**Ans: C**

From energy conservation:

$$\frac{1}{2}mv^2 = mgh$$

$$v^2 = 2gh = 2gr$$

At the bottom:

$$N - mg = \frac{mv^2}{r}$$

$$N = mg + \frac{mv^2}{r} = mg + 2mg = 3mg$$

- 10** Taking the Earth to be a perfect sphere of uniform density rotating about its polar axis,